

Scenario 1

32 YOF w/ Pulmonary Embolism

Scenario Set Up	Equipment: Blood Pressure Cuff, Stethoscope, Pulse Oximeter, NRB, O2 PROCTOR: Patient cannot speak in detail, is constantly trying to catch their breath. The second FTO is the husband of the patient and is very anxious about his wife's condition.
Dispatch	You are dispatched to a 32-year-old female experiencing shortness of breath and difficulty breathing.
Scene Size Up	Scene Safety: No hazards, scene safe, one patient, additional resources if needed MOI/NOI: Nature of Illness (shortness of breath) General Impression: Young adult female with moderate respiratory distress, visibly anxious, sitting on the couch with her hand on her chest feeling pain Husband present and anxious about his wife's condition
Pertinent Primary Assessment Findings	Patient is alert and tracks EMTs (A&O x4) A - airway patent B - rapid and shallow, no wheezing, no crackles (RR: 30/min, SpO2: 85%, lung sounds clear bilaterally) C - rapid radial present, skin pale, cool, and diaphoretic, cap refill: 3 sec (HR: 124/min)

Pertinent Secondary Assessment Findings	A - shell fish M - LexaPro (anxiety) Sprintec (birth control) P - diagnosed with anxiety, no known cardiac or respiratory disease history, no history of asthma, no previous pulmonary embolism L - ate taco bell 3 hours ago E - was walking from the bathroom to the living room and suddenly couldn't catch her breath, symptoms became worse over 20 min O - happened approx 20 min ago P - breathing deeply makes the chest pain worse, sitting upright makes it a little better Q - chest pain is sharp and stabbing R - right-sided chest pain, no radiation S - 7/10 T - began 20 mins ago <u>* left calf appear slightly swollen, pt reports left calf has been sore for a few days</u> <u>→ when asked further, husband responds they just got back from an international flight visiting their daughter in Australia</u>
Vitals	<u>Initial</u> HR 124 bpm BP 108/68 RR 30/min SpO2 85% Skin - pale, cool, diaphoretic

	Temp 99 <u>After Oxygen</u> HR 118 bpm BP 106/66 RR 28/min SpO2 95% Skin - still pale/diaphoretic
Treatments	Place patient in a position of comfort, administer O2 via NRB at 15 L/min, rapid transport/consider requesting ALS, reassess every 5 mins
Key Points	Rapid Transport, cannot administer any medications for the chest pain, important to place patient in position of comfort *important to notice that lung sounds are normal even with significant respiratory distress Differential Diagnosis: Deep Vein Thrombosis secondary to travel → Pulmonary Embolism
Bonus Questions	What are some risk factors for developing a pulmonary embolism? <i>Blood clotting disorders via genetic mutations, travel, recent surgery/trauma</i> What type of shock can occur with a pulmonary embolism? <i>Obstructive shock (pulmonary arteries)</i> What is the difference between respiratory distress, respiratory failure, and respiratory arrest? <i>Distress: increased work of breathing</i> <i>Failure: blood is inadequately oxygenated</i> <i>Arrest: no breathing</i>

Scenario 2

26 YO M w/ Tension Pneomothorax

Scenario Set Up	<i>Equipment:</i> Trauma shears, vented and unvented occlusive dressings, BVM and oxygen, stethoscope, BP cuff, pulse oximeter <i>PROCTOR: (Patient and friend)</i> Patient's hunting partner is very panicked and distressed. You are in the woods and accidentally tripped and shot your friend's chest, called 911. PT lying flat, restless, extreme pain, conscious, shallow breathing.
Dispatch	<i>Hunting accident at a trailhead. Caller states a 26 YOM has suffered a gunshot wound to the chest. Law enforcement has secured the scene. Clear to enter</i>
Scene Size Up	All safe, no hazards, 1 patient, mechanism of injury, need for ALS, no need for extrication 26 YOM supine in dirt, friend is next to him in distress. growing pool of blood underneath PT. small cut on forehead.
Pertinent Primary Assessment Findings	<i>AVPU - V</i> <i>A&O X 1 (only to name)</i> X - growing pool of blood underneath PT A - Patent B - rapid ~30b/m, shallow, extremely labored, unequal chest rise, right side does not expand. (Absent lung sounds of right side if they ask) C - Pulse: weak, thready, rapid <i>Skin: pale cool diaphetic,</i> <i>Capillary refill: >4s</i> <i>Obvious bleeding: Heavy bleeding back right side. Needs</i>

	<i>vented dressing immediately.</i>
Pertinent Secondary Assessment Findings	<i>If they try to put a vent on the front bullet wound, it slides off and wont stick because of blood and sweat. They need to wipe it, then apply it for it to stick.</i> From Friend: A: None M: Ibuprofen sometimes P: Nothing L: beef jerky and water 1 hour ago E: I was walking behind him, caught my boot on a root, and fell. My rifle went off and hit him in the back/shoulder area. He dropped immediately and couldn't breathe O: suddenly, but has gotten slowly worse P: Breathing or moving Q: unknown R: unknown S: Unknown T: 20 minutes ago Trauma: During trauma, the friend tries to stop you from ruining his clothes. Forehead scrape with minimal bleeding <u>Trachea deviated to left</u> <u>Jugular Vein distention</u> <u>Absent lung sounds right side</u> <u>Sucking chest wound, on right side after exposed, front and back</u>
Vitals	<i>Initial:</i> <i>HR:132</i> <i>BP: 88/50</i> <i>RR:30, shallow, labored</i> <i>Sugar: 105 mg/dl</i> <i>SPO2:84% on room air</i> <i>Eyes: PERRL</i> <i>Second: (after vented chest seal and O2)</i> <i>HR: 134</i>

	<i>BP: 96/55</i> <i>RR: 28, less shallow, less labored</i> <i>SPO2: 91% w/ BVM 15LPM</i>
Treatments	<i>Immediate:</i> <i>Log roll to address bleeding, pressure then vented seal. Vented chest seal on front too.</i> <i>High flow BVM / NRB</i> <i>Treat for shock: Supine, thermal blanket, rapid transport</i> <i>Non life threatening:</i> <i>Bandage to forehead</i>
Key Points	They must first log roll and do a blood sweep to find major bleeding on back. Do not let friend interfere. Signs of tension pneumothorax: absent lung sounds, JVD, hypotension, tracheal deviation. BVM can worsen pneumothorax, but in this case the pt cannot maintain his respirations so there is no other choice. Do not hyperventilate.
Bonus Questions	What is the definitive field treatment for a tension pneumothorax, and is it within the EMT-Basic scope of practice? <i>- Needle thoracostomy, ALS only. Why rapid transport or ALS is essential</i> If you are ventilating this patient with a BVM, what physical sensation might you notice if the tension pneumothorax worsens? <i>- Increased BVM resistance</i> Why does blood pressure drop in tension pneumothorax? <i>- Vena cava compressed -> less blood returned to the heart</i>

Scenario 3

19 YO Female with Asthma

Scenario Set Up	<i>Equipment: Stethoscope, BP cuff, Pulse Ox, Inhaler (prescribed to patient and on them),</i> <i>PROCTOR: Patient is a 19-year-old female with a history of asthma. She is at her college dorm after developing progressively worsening difficulty breathing. A roommate called 911. The patient should be sitting upright, leaning forward, anxious, and visibly struggling to breathe. She should initially be able to answer questions but only in short phrases. Have the patient become increasingly fatigued later in the scenario.</i> <i>Bystander/roommate is present and can provide additional history if needed.</i>
Dispatch	<i>19-year-old female experiencing difficulty breathing at a college dorm. Roommate called 911. Unknown if the patient has taken any medication.</i>
Scene Size Up	<i>Scene is safe. See patient sitting on ground having difficulty breathing and is wheezing. Patient appears anxious and is using accessory muscles to breathe. No obvious trauma or bleeding. Roommate is present. No immediate environmental hazards.</i>
Pertinent Primary Assessment Findings	<i>AVPU - A</i> <i>A&O X 4</i> <i>A - patent</i> <i>B - Rapid labored, 34 RR, can only speak in 2-3 word sentences, wheezing, 86% spO2</i> <i>C - Radial pulse present, rapid and regular. HR: 128 bpm Skin is pale, cool, and diaphoretic. Capillary refill under 2 seconds.</i>

Pertinent Secondary Assessment Findings	<i>P - Started around 10-15 mins ago</i> <i>A - No associated chest pain just feels tight</i> <i>S - No sputum</i> <i>T - 2-3 words per sentence</i> <i>E - Can stand but walking to gurney is difficult</i> <i>A - No</i> <i>M - Albuterol</i> <i>P - Asthma</i> <i>L - Lunch 2 hours ago</i> <i>E - I started feeling tight in my chest and wheezing about 30 minutes ago. It kept getting worse. I tried walking around and using my inhaler, but I'm still having trouble breathing.</i>
Vitals	BP: 142/88 , HR:128, RR:36, BGL:108, SPO2:86% <i>After NRB within around 10 mins:</i> BP: 132/88 , HR:124, RR:30, BGL:108, SPO2:92% <i>After some time patient starts deteriorating</i> <i>New findings:</i> • RR: 10/min, HR: 138 bpm SpO₂: 78% - patient is leading towards respiratory failure - rapid transport
Treatments	<i>NRB initially, rapid transport, patient can administer their own albuterol</i>
Key Points	<i>High flow O2, Assist with inhaler</i>
Bonus Questions	Can you administer albuterol? <i>No, can only assist/encourage pt to use it as per ALCO protocol</i> Are there any other interventions we can do as EMTs in this situation? <i>Watch for signs of anaphylaxis (epiPen)</i>